Spectroscopic Techniques

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Spectroscopic Techniques (A29283)

Short Title:	Spectroscopic Techniques		
Faculty	Science and Computing		
Department	Science		
Cluster	Analytical Science		
Author	CWALSH		
Credits	5	**Level:**	Introductory

Description of Module / Aims

This module introduces the learner to a variety of spectroscopic techniques encountered routinely in the analytical laboratory. The principles of operation of Ultraviolet-Visible (UV-Vis), Infra-red (IR) and Atomic Absorption Spectroscopy (AAS) will be described, along with their associated optical components and instrumentation. This module will also cover physical optics associated with spectroscopy, including the origin of spectra. Laboratory practicals will enable the student to gain hands-on experience of using these techniques to perform qualitative and quantitative analyses on a range of samples.

Programmes

`SCIE-0040`	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (`WD_SMBIO_B`)	2 / 3 / M
`SCIE-0040`	BSc (Hons) in Pharmaceutical Science (`WD_SPHSC_B`)	2 / 3 / M
`SCIE-0040`	BSc (Hons) in Physics for Modern Technology (`WD_KPHTE_B`)	2 / 3 / M
`SCIE-0040`	BSc in Molecular Biology with Biopharmaceutical Science (`WD_SMBIO_D`)	2 / 3 / M
`SCIE-0040`	BSc in Pharmaceutical Science (`WD_SPHSC_D`)	2 / 3 / M

Indicative Content

- Review of nature of electromagnetic waves and properties of the electromagnetic spectrum
- Principles of diffraction, superposition, interference and coherence of electromagnetic waves, overview of Michelson interferometer
- Atomic spectra, energy levels, electron orbits, wave-particle duality, matter waves, energy quantisation
- Principle of operation of a number of photodetectors: photodiode, photomultiplier tube
- Introduction to the principles, instrumentation and experimental practices of UV-Vis spectroscopy, including quantitative analysis using the Beer-Lambert Law
- Introduction to the principles, instrumentation and experimental practices of Infrared (IR) spectroscopy, including qualitative analysis in organic molecules
- Introduction to the principles, instrumentation and experimental practices of Atomic Absorption Spectroscopy (AAS), including quantitative analysis of metals
- Associated laboratory-based experiments to supplement and augment the above content, reinforcing good laboratory practices

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Discuss the nature of the interaction between electromagnetic radiation and matter.
2. Describe the phenomena associated with physical optics and the principle of operation of photodetectors.
3. Explain the origin of energy quantisation in atomic systems.
4. Describe the principles of operation and associated instrumentation of the following spectroscopic techniques: UV-Vis, IR and AAS.
5. Apply the laws of spectroscopy to perform a quantitative analysis on a range of samples using UV-Vis and AAS.
6. Apply IR spectroscopy for the qualitative analysis of a range of samples, including interpreting spectra.
7. Demonstrate good laboratory practices with spectroscopic instrumentation and data analysis.

Learning and Teaching Methods

- Lectures.
- Practicals.

Assessment Methods

	Weighing		Outcomes Assessed
Final Written Examination	50%		1,2,3,4
Continuous Assessment	50%		
Practical		50%	5,6,7

Assessment Criteria

<40%: The learner will demonstrate little or no understanding of the fundamentals of the learning outcomes. Will display a lack of competence with regards to associated laboratory skills, use of instrumentation and data analysis.

40%-49%: The learner will demonstrate limited understanding of the fundamentals of the learning outcomes. Will display some ability with regards to associated laboratory skills, use of instrumentation and data analysis.

50%-59%: As well as the above, the learner will demonstrate reasonable understanding of the fundamentals of the learning outcomes. Will display reasonable competence with regards to associated laboratory skills, use of instrumentation and data analysis, combined with good laboratory practice.

60%-69%: In addition, the learner will demonstrate more detailed knowledge of the learning outcomes. Will display good competence with regards to associated laboratory skills, use of instrumentation and data analysis, combined with a very good level of good laboratory practice.

70%-100%: All of the above to excellent levels. In addition, the learner will demonstrate comprehensive knowledge and understanding of all learning outcomes. Will display advanced competence and independence with regards to associated laboratory skills, use of instrumentation and data analysis, combined with an excellent level of good laboratory practice.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	36	
Independent Learning	75	

Essential Material

- "Spectroscopic Techniques module Moodle page." http://moodle.wit.ie
- Skoog, D.A., J. Holler and S.R. Crouch. _Principles of Instrumental Analysis_. 6th ed. London: Thomson Learning, 2007.
- Walker, J. S. _Physics Technology Update_. 4th ed. United States of America: Pearson, 2014.

Supplementary Material

- Khopkar, S.M. _Basic Concepts of Analytical Chemistry_. 3rd ed.. Kent: New Academic Science, 2012.

Requested Resources

- Room Type: Chemistry Lab
- SCIENCE LAB: Physics